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Problem statement:Read the marks obtained by students of second year in an online examination of particular subject. Find out maximum and minimum marks obtained in that subject. Use heap data structure. Analyze the algorithm.

```
import heapq

def find_min_max_marks(marks):
    # Create a min-heap and max-heap
    min_heap = []
    max_heap = []

    # Insert all marks into both heaps
    for mark in marks:
        heapq.heappush(min_heap, mark)  # Min-heap (default behavior of
heapq)
        heapq.heappush(max_heap, -mark)  # Max-heap (invert the sign to
use min-heap as max-heap)

    # The minimum mark will be the root of the min-heap
    min_mark = min_heap[0]

    # The maximum mark will be the root of the max-heap
    max_mark = -max_heap[0]

    return min_mark, max_mark

# Example marks of students
marks = [85, 92, 78, 65, 88, 79, 95, 82, 70]

min_mark, max_mark = find_min_max_marks(marks)

print(f"Minimum mark: {min_mark}")
print(f"Maximum mark: {max_mark}")
```

Output :

```
Minimum mark: 65
Maximum mark: 95
```